Waves at boundaries

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Describe reflection and transmission at a boundary.

SUCCESS CRITERIA - I can...

1. Identify incident, reflected and transmitted waves.
2. Distinguish fixed-end from free-end pulse reflection.
3. Track energy without assuming equal amplitudes.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “boundary” and “reflection” to explain your model or result. Sentence starters are optional.

boundary: the place where a medium or constraint changes

reflection: a wave returning from a boundary

transmission: a wave continuing through or beyond a boundary

01 Bring back an earlier idea

Echoes return at 0.003 s and 0.005 s with the same wave speed.
Compare ranges. Does delay alone determine mass?

Word help: wave = a disturbance that can carry energy and information

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $f_{reflected} = f_{incident}$.
At a stationary boundary. A displacement pulse returns inverted from an ideal fixed end and upright from an ideal free end.

Worked reasoning: At a fixed end, an upward rope pulse reflects downward. At an ideal free end it reflects without inversion. These are different boundary conditions.

Your turn: Draw incident and reflected pulse snapshots for both ends; show propagation directions.

Sentence starter: The ... in my model represents ... . Its label shows ... .

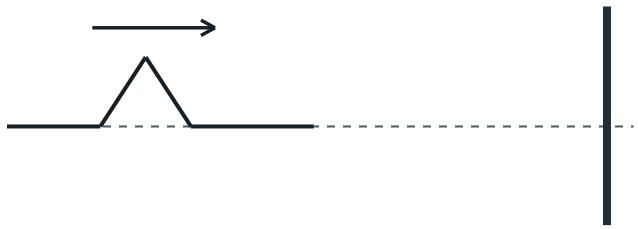
Marzano check - circle: ① ② ③

03 Represent and interpret

An upward rope pulse reaches an ideal fixed end.
Sketch the reflected pulse and its travel direction.

Word help: ideal = using stated simplifying conditions

Sentence starter: The ... in my model represents ... . Its label shows ... .



Sketch and label your response.

Marzano check - circle: ① ② ③

04 Solve and justify

Incident energy is .80 J; reflected energy is .20 J and absorbed energy .10 J.
Find transmitted energy and each fraction.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

Does a smaller transmitted amplitude prove frequency changed?
Explain what usually fixes the frequency at a stationary boundary.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

06 Choose a method

A 0.4 kg liquid releases 120000 J while condensing at constant temperature.
Find latent heat and signed Q for the sample.

Word help: latent heat = energy per mass needed for a specified phase change

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

07 Exit check

At a stationary boundary frequency stays 5 Hz while wave speed changes from 10 to 20 m/s.
Find wavelength on each side.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A pulse reflects from a fixed end and meets an equal incident pulse. At an instant their displacements are +2 and -2 cm at one point.
Find total displacement and explain why zero there does not mean all energy vanished.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: 1 2 3